

Foglio 2 - Soluzioni,

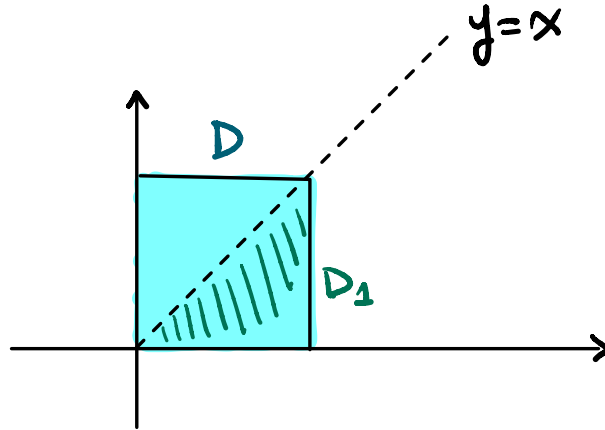
ESERCIZIO 1 (Diario del corso, es. 31)

Calcolare

$$\iint_D \sqrt{x^2+y^2} \, dx \, dy$$

con $D = [0,1] \times [0,1]$.

Soluzioni



Neliamo che D è simmetrico rispetto alla retta $y=x$.

Inoltre, posto $f = \sqrt{x^2+y^2}$, si ha $f(x,y) = f(y,x)$

Pertanto, $\iint_D \sqrt{x^2+y^2} \, dx \, dy = 2 \iint_{D_1} \sqrt{x^2+y^2} \, dx \, dy =: I$

Descriviamo D_1 in polari. ($x = \rho \cos \theta$, $y = \rho \sin \theta$).

Chiaramente, $\theta \in [0, \pi/4]$.

$$\text{Poi } \begin{cases} 0 \leq \rho \cos \theta \leq 1 \\ 0 \leq \rho \sin \theta \leq 1 \end{cases} \implies 0 \leq \rho \leq \min \left\{ \frac{1}{\cos \theta}, \frac{1}{\sin \theta} \right\} \overset{\theta \in [0, \pi/4]}{=} \frac{1}{\cos \theta}$$

D_1 è dunque θ -normale

$$I = 2 \int_0^{\pi/4} \left(\int_0^{1/\cos\theta} p^2 dp \right) d\theta$$

$$= \frac{2}{3} \int_0^{\pi/4} \frac{1}{\cos^3\theta} d\theta$$

$$= \frac{2}{3} \int_0^{\pi/4} \frac{1}{\cos\theta (1-\sin^2\theta)} d\theta = \frac{2}{3} \int_0^{\pi/4} \frac{\cos\theta}{\cos^2\theta (1-\sin^2\theta)} d\theta$$

$$\begin{array}{l} \sin\theta = u \\ \cos\theta d\theta = du \end{array} = \frac{2}{3} \int_0^{1/\sqrt{2}} \frac{du}{(1-u^2)^2} = (*)$$

$$\frac{1}{(1-u^2)^2} = \frac{1}{(1-u)^2(1+u)^2} \underset{\substack{\uparrow \\ \text{fatti semplici}}}{=} \frac{A}{(1-u)} + \frac{B}{(1-u)^2} + \frac{C}{(1+u)} + \frac{D}{(1+u)^2} \overset{\text{conti}}{\rightsquigarrow} A=B=C=D=1/4$$

$$\begin{aligned} (*) &= \frac{1}{4} \int_0^{1/\sqrt{2}} \frac{1}{(1-u)} du + \frac{1}{4} \int_0^{1/\sqrt{2}} \frac{1}{(1+u)} du + \frac{1}{4} \int_0^{1/\sqrt{2}} \frac{1}{(1-u)^2} du + \frac{1}{4} \int_0^{1/\sqrt{2}} \frac{1}{(1+u)^2} du \\ &= -\frac{1}{4} \log(1-u) \Big|_0^{1/\sqrt{2}} + \frac{1}{4} \log(1+u) \Big|_0^{1/\sqrt{2}} + \frac{1}{4} \frac{1}{(1-u)} \Big|_0^{1/\sqrt{2}} - \frac{1}{4} \frac{1}{(1+u)} \Big|_0^{1/\sqrt{2}} \\ &= -\frac{1}{6} \log\left(\frac{2-\sqrt{2}}{2+\sqrt{2}}\right) + \frac{\sqrt{2}}{3} \end{aligned}$$

ESERCIZIO 2

Calcolare

$$\iint_E \frac{(y^2 - x^2) e^{(x-y)^2 + \sqrt{xy}}}{\sqrt{xy}} dx dy$$

con $E = \{(x, y) \in \mathbb{R}^2 \mid 1 \leq xy \leq 2, x \leq y \leq x+1, x, y \geq 0\}$

Soluzione

Poniamo $f(x, y, z) = \iint_E \frac{(y^2 - x^2) e^{(x-y)^2 + \sqrt{xy}}}{\sqrt{xy}} dx dy$. $E: \begin{cases} 1/x \leq y \leq 2/x \\ x \leq y \leq x+1 \end{cases}$

Usiamo il cambio di coordinate $T: \mathbb{R}^2 \xrightarrow{e^\infty} \mathbb{R}^2$

$$(u, v) \mapsto \left(\underbrace{\frac{u-v}{2}}_{x(u,v)}, \underbrace{\frac{u+v}{2}}_{y(u,v)} \right)$$

$$|\det J_T(u, v)| = 1/2 \leadsto dx dy = 1/2 du dv$$

$$xy = \frac{(u-v)}{2} \cdot \frac{(u+v)}{2} = \frac{u^2 - v^2}{4}$$

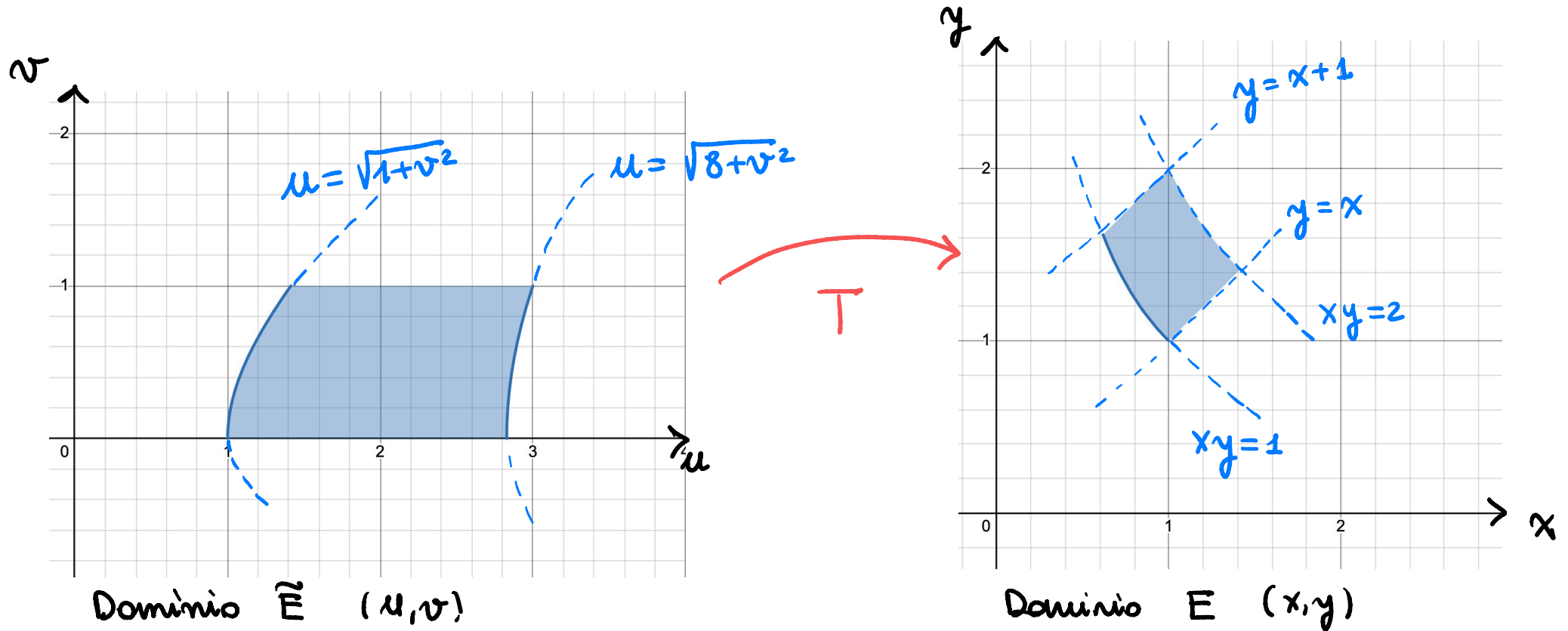
Dunque: $x \leq y \leq x+1 \iff 0 \leq y-x \leq 1 \iff 0 \leq v \leq 1$

$$1 \leq xy \leq 2 \iff 1 \leq \frac{u^2 - v^2}{4} \leq 2 \iff 4 + v^2 \leq u^2 \leq 8 + v^2$$

Notiamo poi che su $\tilde{E}$ $u \geq 0$ perché $u = x + y$ e per $(x, y) \in E$ si ha $x, y \geq 0$.
 Pertanto $\sqrt{4+v^2} \leq u \leq \sqrt{8+v^2}$.

Riassumendo: il dominio E , si scrive rispetto alle coordinate (u, v) come

$$T^{-1}(E) =: \tilde{E} = \{(u, v) \mid 0 \leq v \leq 1, \sqrt{4+v^2} \leq u \leq \sqrt{8+v^2}\} \rightarrow \bar{e} \text{ } v\text{-normale!}$$



Poi:

$$f(x(u,v), y(u,v)) = \frac{uve^{v^2 + \sqrt{\frac{u^2-v^2}{4}}}}{\sqrt{\frac{u^2-v^2}{4}}}$$

Finally:

$$\iint_E f(x,y) dx dy = \iint_E \frac{uve^{v^2 + \sqrt{\frac{u^2-v^2}{4}}}}{\sqrt{\frac{u^2-v^2}{4}}} \frac{1}{2} du dv$$

$$= \int_0^1 \left(ve^{v^2} \int_{\sqrt{4+v^2}}^{\sqrt{8+v^2}} \frac{ue^{\sqrt{\frac{u^2-v^2}{4}}}}{\sqrt{u^2-v^2}} du \right) dv$$

$$= \int_0^1 \left(2ve^{v^2} \left[e^{\sqrt{\frac{u^2-v^2}{4}}} \right]_{u=\sqrt{4+v^2}}^{u=\sqrt{8+v^2}} \right) dv$$

$$= (e^{\sqrt{2}} - e) \int_0^1 2ve^{v^2} dv = (e^{\sqrt{2}} - e) e^{v^2} \Big|_0^1 = (e^{\sqrt{2}} - e)(e - 1)$$

$$\frac{\partial}{\partial u} \left(e^{\sqrt{\frac{u^2-v^2}{4}}} \right) = \frac{ue^{\sqrt{\frac{u^2-v^2}{4}}}}{2\sqrt{u^2-v^2}}$$

ESERCIZIO 3 (Diario del Corso, es. 17)

Considera il corpo $D = \{(x, y) \in \mathbb{R}^2 \mid x^2/4 + y^2 \leq 1\}$, avente densità di massa

$$\mu(x, y) = \begin{cases} 1 & \text{se } x \leq 0 \\ 1+x^2 & \text{se } x \geq 0 \end{cases}$$

Calcolare il centro di massa di D .

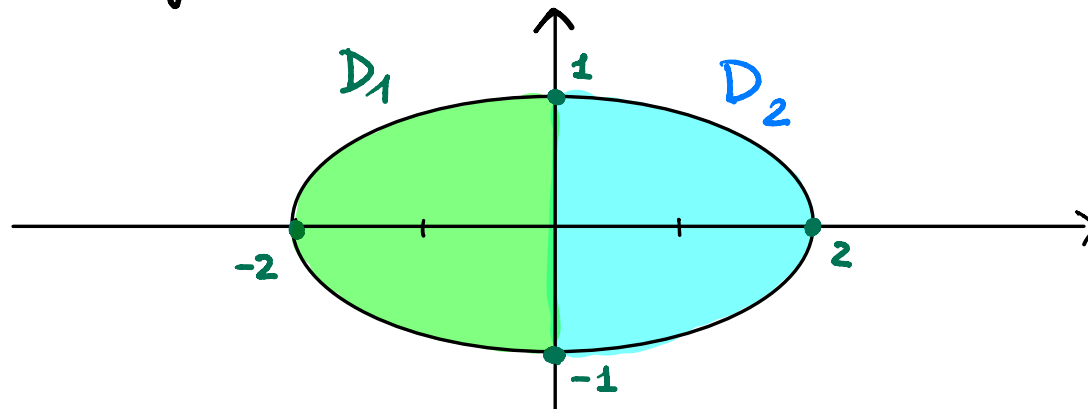
Soluzione

Il centro di massa $\bar{x}$ dato da $CM = (x_{cm}, y_{cm})$ con:

$$\begin{cases} x_{cm} = \frac{1}{\text{massa}(D)} \iint_D x \mu(x, y) dx dy \\ y_{cm} = \frac{1}{\text{massa}(D)} \iint_D y \mu(x, y) dx dy \end{cases}$$

Partiamo da

$$\text{massa}(D) = \iint_D \mu dx dy.$$



Dividiamo D in due parti, $D = D_1 \cup D_2$

$$D_1 = \left\{ -2 \leq x \leq 0 \mid -\sqrt{1-\frac{x^2}{4}} \leq y \leq \sqrt{1-\frac{x^2}{4}} \right\}$$

$$D_2 = \left\{ 0 \leq x \leq 2 \mid -\sqrt{1-\frac{x^2}{4}} \leq y \leq \sqrt{1-\frac{x^2}{4}} \right\}$$

$$\begin{aligned} \text{massa}(D) &= \iint_{D_1} dx dy + \iint_{D_2} (x^2 + 1) dx dy \\ &= \iint_{D_1} dx dy + \iint_{D_2} dx dy + \iint_{D_2} x^2 dx dy \\ &= \text{Area}(D) + \iint_{D_2} x^2 dx dy \end{aligned}$$

$\text{Area}(D) = \text{area ellisse semiassi } 2 \text{ e } 1 = \pi \cdot 2 \cdot 1 = 2\pi.$

Resta da calcolare $\iint_{D_2} x^2 dx dy$ simmetria dom + f_z pari rispetto ad y

dove $D_2^+ = D_2 \cap \{y \geq 0\} = \left\{ 0 \leq x \leq 2 \mid 0 \leq y \leq \sqrt{1-\frac{x^2}{4}} \right\}.$

$$\iint_{D_2} x^2 dx dy = 2 \iint_{D_2^+} x^2 dx dy$$

$$= 2 \int_0^2 \left(\int_0^{\sqrt{1-x^2/4}} x^2 dy \right) dx$$

$$= 2 \int_0^2 x^2 \sqrt{1-x^2/4} dx$$

$$= 4 \int_0^{\pi/2} 4 \sin^2 z \sqrt{1-\sin^2 z} \cos z dz$$

$$= 16 \int_0^{\pi/2} \sin^2 z \cos^2 z dz$$

$$= 16 \int_0^{\pi/2} \frac{\sin^2(2z)}{4} dz$$

$$= 4 \int_0^{\pi/2} \sin^2(2z) dz$$

bisezione

$$\sin^2(2z) = \frac{1 - \cos 4z}{2}$$

$$= 2 \int_0^{\pi/2} dz - \frac{2}{4} \int_0^{\pi/2} 4 \cos(4z) dz$$

$$= \pi - \frac{1}{2} \left[-\sin(4z) \right]_0^{\pi/2} = \pi$$

~> massa (D) = 3π

$$x = 2 \sin z$$

$$z = \arcsin \frac{x}{2}$$

$$dx = 2 \cos z dz$$

Ora calcoliamo

$$\begin{aligned} x_{cm} &= \frac{1}{\text{massa}(D)} \iint_D x \rho(x,y) dx dy \\ &= \frac{1}{3\pi} \left[\iint_{D_1} x dx dy + \iint_{D_2} (x + x^3) dx dy \right] \\ &= \frac{1}{3\pi} \left[\underbrace{\iint_D x dx dy}_0 + \iint_{D_2} x^3 dx dy \right] \end{aligned}$$

per simmetria!

Resta da calcolare

simmetria dominio + funzione pari rispetto a y

$$\begin{aligned} \iint_{D_2} x^3 dx dy &= 2 \iint_{D_2^+} x^3 dx dy \\ &= 2 \int_0^2 \left(\int_0^{\sqrt{1-x^2/4}} x^3 dy \right) dx \\ &= 2 \int_0^2 x^3 \sqrt{1 - \frac{x^2}{4}} dx \\ &= \int_0^2 x^3 \sqrt{4 - x^2} dx \end{aligned}$$

$$\begin{aligned}
 4-x^2=u & \\
 -2x dx = du & \\
 &= \int_4^0 \left(\frac{u-4}{2} \right) \sqrt{u} du \\
 &= -\frac{1}{2} \int_0^4 (u^{3/2} - 4u^{1/2}) du \\
 &= -\frac{1}{2} \left[\frac{2}{5} u^{5/2} - \frac{8}{3} u^{3/2} \right]_0^4 \\
 &= -\frac{1}{2} \left(\frac{2}{5} \cdot 2^5 - \frac{8}{3} 2^3 \right) \\
 &= -\frac{1}{2} \left(\frac{64}{5} - \frac{64}{3} \right) = -\frac{1}{2} \cdot \left(-\frac{192-320}{15} \right) = \frac{128}{30} = \frac{64}{15}
 \end{aligned}$$

$$\leadsto x_{cm} = 64\pi/45$$

$$\begin{aligned}
 y_{cm} &= \frac{1}{\text{massa}(D)} \iint_D y \mu(x,y) dx dy \\
 &= \frac{1}{3\pi} \left[\iint_{D_1} y dx dy + \iint_{D_2} (y + x^2 y) dx dy \right] \\
 &= \frac{1}{3\pi} \left[\underbrace{\iint_D y dx dy}_{} + \underbrace{\iint_{D_2} x^2 y dx dy}_{} \right] = 0
 \end{aligned}$$

||
0

||
0

funzione y -dispari
+
dominio simmetrico
rispetto a $ne\ x$

$$CM = \left(\frac{64}{45}\pi, 0 \right)$$

ESERCIZIO 4

Calcolare

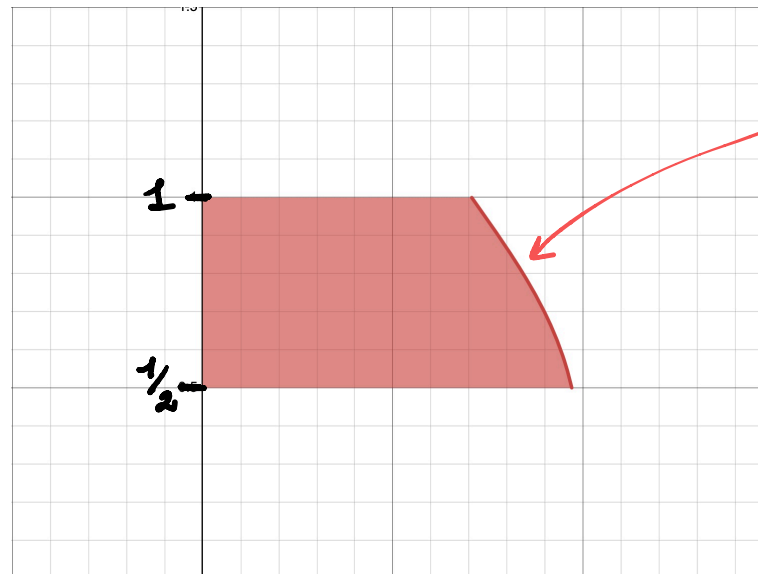
$$\iint_H \frac{\cos^2(\arctan y^2)}{\sqrt{x}} dx dy$$

$$\text{dove } H = \{(x, y) \in \mathbb{R}^2 \mid x^2 - \frac{1}{1+y^4} \leq 0, x \geq 0, |y - \frac{3}{4}| \leq \frac{1}{4}\}$$

Soluzione

Il dominio è y -normale, scrivibile come

$$H = \{(x, y) \in \mathbb{R}^2 \mid \frac{1}{2} \leq y \leq 1, 0 \leq x \leq \frac{1}{\sqrt{1+y^4}}\}$$



$$x = \frac{1}{\sqrt{1+y^4}}$$

Integriamo.

$$\begin{aligned}
 \iint_H \frac{\cos^2(\arctan y^2)}{\sqrt{x}} dx dy &= \int_{1/2}^1 \left(\int_0^{1/\sqrt{1+y^4}} \frac{\cos^2(\arctan y^2)}{\sqrt{x}} dx \right. \\
 &= \int_{1/2}^1 \left(\cos^2(\arctan y^2) \cdot \int_0^{1/\sqrt{1+y^4}} \frac{1}{\sqrt{x}} dx \right) dy \\
 &= 2 \int_{1/2}^1 \frac{\cos^2(\arctan y^2)}{\sqrt[4]{1+y^4}} dy \\
 &\xrightarrow{\substack{u = \arctan y^2 \\ y = \sqrt{\tan u} \\ dy = \frac{1}{2\sqrt{\tan u}} \cdot \frac{1}{\cos^2 u} du \\ \sqrt[4]{1+y^4} = \frac{1}{\sqrt{\cos u}}}} 2 \int_{\arctan(1/4)}^{\pi/4} \frac{\cancel{\cos^2 u} \sqrt{\cos u}}{2 \sqrt{\tan u}} \cdot \frac{1}{\cancel{\cos^2 u}} du \\
 &= \int_{\arctan(1/4)}^{\pi/4} \frac{\cos u}{\sqrt{\sin u}} du \\
 &= 2 \left[\sin^{1/2} u \right]_{\arctan(1/4)}^{\pi/4} \\
 &= 2 \left[\frac{1}{\sqrt[4]{2}} - \sqrt{\sin(\arctan(1/4))} \right] = \frac{1}{2} \left(\frac{1}{\sqrt[4]{2}} - \frac{1}{\sqrt[4]{17}} \right)
 \end{aligned}$$

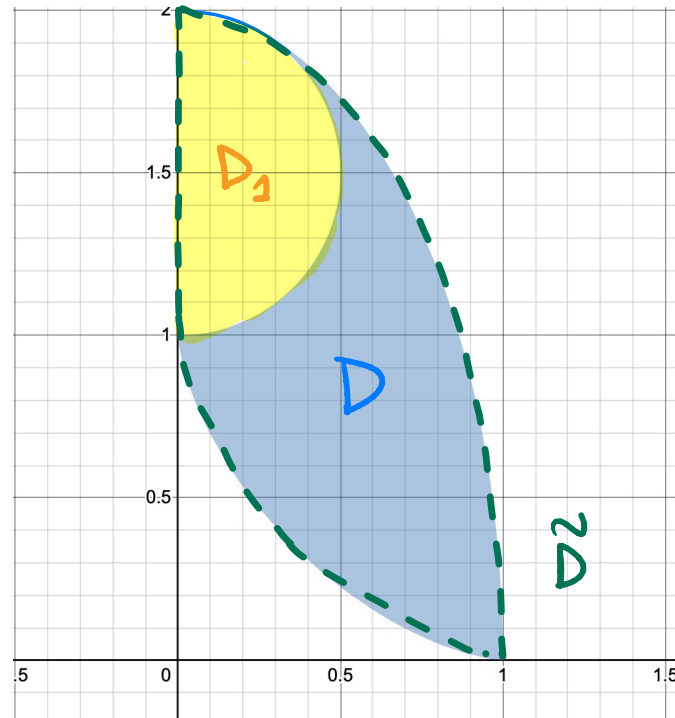
$$\sin(\arctan(x)) = \frac{x}{\sqrt{1+x^2}}$$

ESERCIZIO 5 (Scritto 21/07/2025)

Sia $D = \{(x, y) \in \mathbb{R}^2 \mid x \geq 0, y \geq 0, x^{2/3} + y^{2/3} \geq 1, x^2 + y^2/4 \leq 1, x^2 + (y - \frac{3}{2})^2 \geq \frac{1}{4}\}$.

Calcolare $\text{Area}(D)$.

Soluzione



D è l'insieme blu in figura. $\text{Area}(D) = \text{Area}(\tilde{D}) - \text{Area}(D_1)$

$\tilde{D}$ è il dominio tratteggiato in verde. È x -semplice:

$$\tilde{D} = \{(x, y) \in \mathbb{R}^2 \mid 0 \leq x \leq 1, (1 - x^{2/3})^{3/2} \leq y \leq 2\sqrt{1 - x^2}\}$$

D_1 è metà cerchio di raggio $1/2$, ie $\text{Area}(D_1) = \frac{1}{2} \cdot (\frac{1}{2})^2 \pi = \frac{\pi}{8}$.

Resta da calcolare

$$\begin{aligned}\text{Area}(\tilde{D}) &= \iint_{\tilde{D}} dx dy \\ &= \int_0^1 \left(\int_{(1-x^{2/3})^{3/2}}^{2\sqrt{1-x^2}} dy \right) dx \\ &= \int_0^1 2\sqrt{1-x^2} dx - \int_0^1 (1-x^{2/3})^{3/2} dx \\ &= 2 \underbrace{\int_0^1 \sqrt{1-x^2} dx}_{\substack{\text{è area di } 1/4 \\ \text{di archio di} \\ \text{raggio 1!}}} - \int_0^1 (1-x^{2/3})^{3/2} dx \\ &= \frac{\pi}{2} - \int_0^1 (1-x^{2/3})^{3/2} dx\end{aligned}$$

Calcoliamo a parte

$$\begin{aligned}\int_0^1 (1-x^{2/3})^{3/2} dx &= \int_0^{\pi/2} (1-\sin^2(\xi))^{3/2} \cdot 3\sin^2 \xi \cos \xi d\xi \\ &\quad \begin{aligned} x^{1/3} &= \sin \xi, \quad \xi = \arcsin(x^{1/3}) \\ x &= \sin^3 \xi \\ dx &= 3\sin^2 \xi \cos \xi d\xi \end{aligned}\end{aligned}$$

$$\sin^2 \alpha = \frac{1 - \cos(2\alpha)}{2}$$

bisezione

$$\begin{aligned}
 &= 3 \int_0^{\pi/2} \cos^4(\xi) \sin^2(\xi) d\xi \\
 &= \frac{3}{4} \int_0^{\pi/2} \cos^2(\xi) \sin^2(2\xi) d\xi \\
 &= \frac{3}{4} \int_0^{\pi/2} (1 - \sin^2(\xi)) \sin^2(2\xi) d\xi \\
 &= \frac{3}{4} \int_0^{\pi/2} \sin^2(2\xi) d\xi - \frac{3}{4} \int_0^{\pi/2} \sin^2 \xi \sin^2(2\xi) d\xi \\
 &= \frac{3}{4} \int_0^{\pi/2} \sin^2(2\xi) d\xi - \frac{3}{4} \int_0^{\pi/2} \left(\frac{1}{2} - \frac{\cos(2\xi)}{2} \right) \sin^2(2\xi) d\xi \\
 &= \frac{3}{4} \int_0^{\pi/2} \sin^2(2\xi) d\xi - \frac{3}{8} \int_0^{\pi/2} \sin^2(2\xi) d\xi + \frac{3}{8 \cdot 2} \int_0^{\pi/2} 2 \cos(2\xi) \sin^2(2\xi) d\xi \\
 &= \frac{3}{8} \int_0^{\pi/2} \sin^2(2\xi) d\xi + \frac{3}{16} \left[\frac{\sin^3(2\xi)}{3} \right]_0^{\pi/2} \\
 &= \frac{3}{8} \int_0^{\pi/2} \left(\frac{1}{2} - \frac{\cos(4\xi)}{2} \right) d\xi \\
 &= \frac{3}{8} \cdot \frac{\pi}{2} \cdot \frac{1}{2} - \frac{3}{16 \cdot 4} \int_0^{\pi/2} 4 \cos(4\xi) d\xi \\
 &= \frac{3}{32} \pi - \frac{3}{64} \left[\sin(4\xi) \right]_0^{\pi/2} = \frac{3}{32} \pi
 \end{aligned}$$

$$\rightsquigarrow \text{Area}(\tilde{D}) = \pi/2 - 3/32\pi = 13/32\pi$$

$$\text{Finally, } \text{Area}(D) = 13\pi/32 - \pi/8 = 9\pi/32.$$

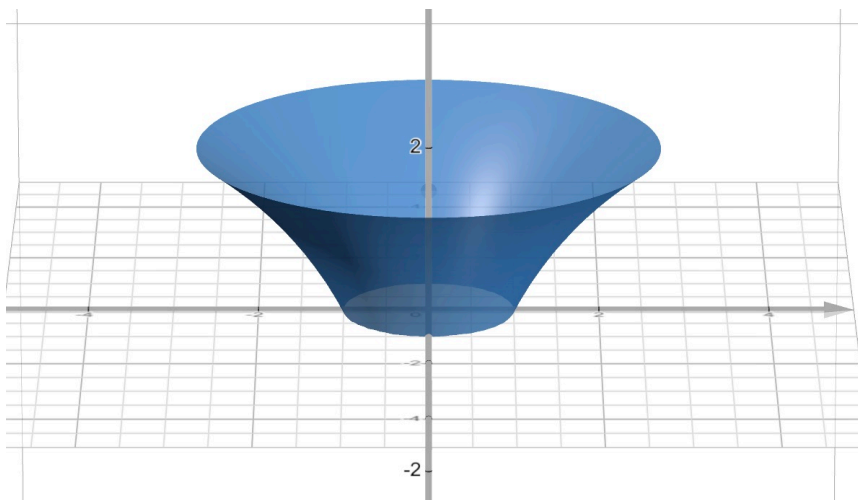
ESERCIZIO 6

Calcolare

$$\int_{\Sigma} \frac{z}{\sqrt{4+x^2+y^2}} d\sigma$$

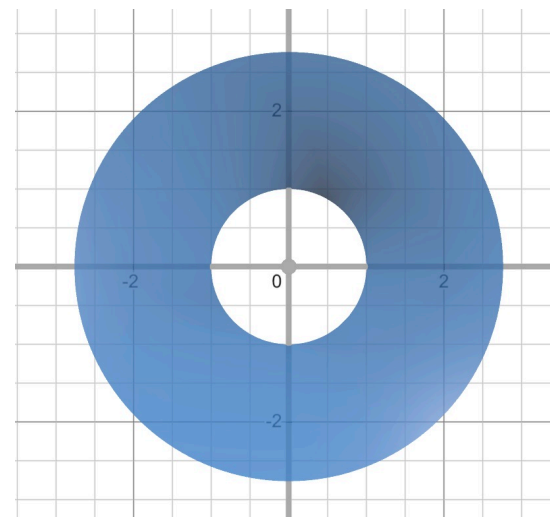
dove $\Sigma = \{(x, y, z) \in \mathbb{R}^3 \mid z = \log(x^2 + y^2), 1 \leq x^2 + y^2 \leq e^2\}$

Soluzione



La superficie Σ è (parte del) grafico di $z = \log(x^2 + y^2)$

Una parametrizzazione di Σ è data da



L'anello $1 \leq x^2 + y^2 \leq e^2$ sul piano $z=0$.

$$\Psi: [1, e] \times [0, 2\pi] \longrightarrow \mathbb{R}^3$$

$$(p, \theta) \longmapsto (p \cos \theta, p \sin \theta, 2 \log p)$$

$$d\sigma = \|\Psi_p \wedge \Psi_\theta\| dp d\theta = \|\Psi_p\| \cdot \|\Psi_\theta\| dp d\theta = p \sqrt{1 + \frac{4}{p^2}}$$

$$\langle \Psi_p, \Psi_\theta \rangle = 0$$

check!

check!

$$\begin{aligned} \int_{\Sigma} \frac{z}{\sqrt{4+x^2+y^2}} d\sigma &= \int_1^e \left(\int_0^{2\pi} \frac{2 \log p}{\sqrt{4+p^2}} \cancel{p \sqrt{1 + \frac{4}{p^2}}} d\theta \right) dp \\ &= 4\pi \int_1^e \log p dp \\ &= 4\pi \left[p \log p - p \right]_1^e = 4\pi \end{aligned}$$